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Hyper-parameter optimization system, method, and program

Abstract

A speech enhancement means **81** determines an enhancement mask generated based on a mask for speech enhancement, when a test utterance is input as speech data. A first hyper-parameter optimization means **82** determines, when the test utterance is input, a first hyper-parameter which is a hyper-parameter representing the degree to which the signal representing the test utterance is kept using the mask, and the first hyper-parameter which is set to take into account a downstream task that is processed using an enhanced test utterance. A mask generation means **83** generates an adaptive mask from the determined enhancement mask and the first hyper-parameter that enhances the test utterance for the downstream task. The mask generation means **83** generates the adaptive mask in which the first hyper-parameter is a power of the mask.

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Background/Summary

(1) This application is a National Stage Entry of PCT/JP2020/038860 filed on Oct. 15, 2020, the contents of all of which are incorporated herein by reference, in their entirety.

TECHNICAL FIELD

(2) The disclosure relates to a hyper-parameter optimization system, hyper-parameter optimization method, and hyper-parameter optimization program for determining an optimal hyper-parameter of masks used for speech enhancement.

BACKGROUND ART

(3) Neural network-based speech enhancement methods have been more promising than general methods that manually turns multiple preprocessing steps. For example, Non-Patent Literature 1 discloses a deep learning based supervised speech separation method as a separation algorithm used when performing speech enhancement.

CITATION LIST

Non Patent Literature

(4) [NPL 1]

(5) D. L. Wang and J. Chen, "Supervised speech separation based on deep learning: An overview", IEEE/ACM, Trans. Audio Speech Lang. Process., 26, pp. 1702-1726, 2018

SUMMARY OF INVENTION

Technical Problem

(6) On the other hand, the purpose of speech enhancement is to improve speech quality. Therefore, speech enhancement does not guarantee subsequent tasks (hereinafter, referred to as downstream tasks) using the enhanced speech, such as speech recognition and speaker recognition. In other words, optimal speech enhancement methods may be different for different downstream tasks. Then, the speech enhancement disclosed in Non-Patent Literature may degrade downstream tasks performance on clean or less noisy speech.

(7) Therefore, it is conceivable to set a mask that enhances the speech according to the downstream task. However, it is difficult for the user to appropriately set hyper-parameters of the mask used for speech enhancement for each downstream task. Therefore, it is preferable that the optimum hyper-parameters of the mask for speech enhancement can be determined according to the nature of the downstream task.

(8) It is an exemplary object of the disclosure to provide a hyper-parameter optimization system, hyper-parameter optimizer method, and hyper-parameter optimization program that can determine the optimum hyper-parameters of the mask that enhances speech according to the nature of the

downstream task.

Solution to Problem

(9) A hyper-parameter optimization system including: a speech enhancement means which determines an enhancement mask generated based on a mask for speech enhancement, when a test utterance is input as speech data; a first hyper-parameter optimization means which determines, when the test utterance is input, a first hyper-parameter which is a hyper-parameter representing the degree to which the signal representing the test utterance is kept using the mask, and the first hyper-parameter which is set to take into account a downstream task that is processed using an enhanced test utterance; and a mask generation means which generates an adaptive mask from the determined enhancement mask and the first hyper-parameter that enhances the test utterance for the downstream task, wherein the mask generation means generates the adaptive mask in which the first hyper-parameter is a power of the mask.

(10) A hyper-parameter optimization method including: determining an enhancement mask generated based on a mask for speech enhancement, when a test utterance is input as speech data; determining, when the test utterance is input, a first hyper-parameter which is a hyper-parameter representing the degree to which the signal representing the test utterance is kept using the mask, and the first hyper-parameter which is set to take into account a downstream task that is processed using an enhanced test utterance; and generating an adaptive mask from the determined enhancement mask and the first hyper-parameter that enhances the test utterance for the downstream task, wherein the adaptive mask in which the first hyper-parameter is a power of the mask is generated.

(11) A hyper-parameter optimization program for causing a computer to execute: a speech enhancement process of determining an enhancement mask generated based on a mask for speech enhancement, when a test utterance is input as speech data; a first hyper-parameter optimization process of determining, when the test utterance is input, a first hyper-parameter which is a hyper-parameter representing the degree to which the signal representing the test utterance is kept using the mask, and the first hyper-parameter which is set to take into account a downstream task that is processed using an enhanced test utterance; and a mask generation process of generating an adaptive mask from the determined enhancement mask and the first hyper-parameter that enhances the test utterance for the downstream task, wherein the adaptive mask in which the first hyper-parameter is a power of the mask is generated.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1

(2) It depicts an exemplary block diagram illustrating the structure of a first exemplary embodiment of a hyper-parameter optimization system according to the disclosure.

(3) FIG. 2

(4) It depicts a flowchart illustrating the process of the first exemplary embodiment of the hyper-parameter optimization system **100**.

(5) FIG. 3

(6) It depicts an exemplary block diagram illustrating the structure of a second exemplary embodiment of a hyper-parameter optimization system according to the disclosure.

(7) FIG. 4

(8) It depicts a flowchart illustrating the process of the second exemplary embodiment of the hyper-parameter optimization system **200**.

(9) FIG. 5

(10) It depicts an exemplary block diagram illustrating the structure of a third exemplary

embodiment of a hyper-parameter optimization system according to the disclosure.

(11) FIG. 6

(12) It depicts a flowchart illustrating the process of the third exemplary embodiment of the hyper-parameter optimization system **300**.

(13) FIG. 7

(14) It depicts an exemplary block diagram illustrating the structure of a fourth exemplary embodiment of a hyper-parameter optimization system according to the disclosure.

(15) FIG. 8

(16) It depicts a flowchart illustrating the process of the fourth exemplary embodiment of the hyper-parameter optimization system **400**.

(17) FIG. 9

(18) It depicts a block diagram illustrating an outline of the hyper-parameter optimization system according to the disclosure.

(19) FIG. 10

(20) It depicts a schematic block diagram illustrating a configuration of a computer according to at least one of the exemplary embodiments.

DESCRIPTION OF EMBODIMENTS

(21) The following describes an exemplary embodiment of the disclosure with reference to drawings.

(22) Note that in the following explanation, when using a Greek letter in the text, an English notation of Greek letter may be enclosed in brackets ([]). Furthermore, the unidirectional arrows shown in each block diagram are a straightforward indication of the direction of information flow and do not exclude bidirectionality.

First Exemplary Embodiment

(23) FIG. 1 depicts an exemplary block diagram illustrating the structure of a first exemplary embodiment of a hyper-parameter optimization system according to the disclosure. The hyper-parameter optimization system **100** according to the first exemplary embodiment includes a training speech input unit **12**, a speech enhancement neural network (hereinafter, it is referred to as speech enhancement NN) parameter storage unit **14**, a first speech enhancement unit **16**, a downstream task neural network (hereinafter, it is referred to as downstream task NN) parameter storage unit **18**, a first hyper-parameter neural network (hereinafter, it is referred to as first hyper-parameter NN) training unit **20**, a first hyper-parameter NN parameter storage unit **22**, a second speech enhancement unit **24**, a first hyper-parameter optimization unit **26**, a mask generation unit **28**, an adapted speech enhancement unit **30**, and a downstream task processing unit **32**.

(24) The training speech input unit **12** accepts speech data for training (hereinafter, it is referred to as training speech) by the first hyper-parameter NN training unit **20**, which will be described below. Specifically, the training speech input unit **12** accepts as a training speech an input of a speech with noise (hereinafter, it is referred to as noisy speech). In this exemplary embodiment, the training speech includes a label indicating a processing result that is assumed to be performed by the downstream task processing unit **32** for the noisy speech (hereinafter, it may be referred to as the downstream task label), as described below.

(25) The noisy speech is created in accordance with the environment (e.g., a situation in which noise is included, a language, a domain, etc.) in which the speech data for the target speech enhancement is acquired. The downstream task label is determined in accordance with the content of the processing performed by the downstream task processing unit **32**. For example, when the downstream task processing unit **32** performs speaker recognition, the downstream task label is a speaker ID or the like.

(26) The training speech input unit **12** may accept training speech input from an external storage server (not shown) and may retrieve the training speech from a storage unit (not shown) provided by the hyper-parameter optimization system **100**.

(27) The speech enhancement NN parameter storage unit **14** stores trained parameters of a neural network that generates an enhancement mask (hereinafter it may be referred to simply as mask), which is generated based on a mask for speech enhancement, from the speech data. The speech enhancement neural network is also referred to as the speech enhancement NN. The enhancement mask is defined, for example, as a power of the hyper-parameter of the mask and is used to enhance the desired speech.

(28) The speech enhancement NN is a trained model for causing a computer to output an enhancement mask from speech data (a speech signal) with noise. Specifically, the speech enhancement NN is a trained neural network that has been machine trained using the training data to calculate an optimal mask (i.e., an enhancement mask) to be used to enhance the desired speech contained in the speech data when speech data with noise is inputted.

(29) The type of the mask used in this exemplary embodiment is not particularly limited. The mask is, for example, a matrix consisting of real or complex continuous values taking at least one form of Ideal Ratio Mask, Complex Ideal Ratio Mask, Spectral Magnitude Mask, and Phase-Sensitive Mask.

(30) In the first exemplary embodiment, it is assumed that the speech enhancement NN is learned beforehand by another learning device (not shown) or the like, and the parameters of the learned speech enhancement NN are stored in the speech enhancement NN parameter storage unit **14**.

(31) The first speech enhancement unit **16** determines a mask (i.e., the enhancement mask) to be used for speech enhancement from the accepted training speech using the speech enhancement NN parameter stored in the speech enhancement NN parameter storage unit **14**. Specifically, the first speech enhancement unit **16** applies the training speech to the input layer of the neural network indicated by the speech enhancement NN parameter, and outputs the enhancement mask from the output layer. Since the first speech enhancement unit **16** obtains the speech enhancement NN parameter stored in the speech enhancement NN parameter storage unit **14**, the first speech enhancement unit **16** can also be said to have a speech enhancement NN.

(32) The downstream task NN parameter storage unit **18** stores parameters of the neural network used by the downstream task processing unit **32** (Hereinafter referred to as the downstream task NN.), which will be described below. In this exemplary embodiment, the downstream task NN parameter storage unit **18** is assumed to store the parameters of the downstream task NN that have already been learned.

(33) The first hyper-parameter NN learning unit **20** learns a neural network (first hyper-parameter NN) in which the first hyper-parameter optimization unit **26**, which will be described below, estimates a hyper-parameter [gamma] (hereinafter referred to as the first hyper-parameter) corresponding to a power of a mask (hereinafter referred to as the adaptive mask) that performs an enhancement adapted to the processing of the downstream tasks on speech data. The hyper-parameter [gamma] is a non-negative scalar value.

(34) This first hyper-parameter is a hyper-parameter that is set to take into account the downstream task and represents the degree to which the signal representing the speech data used for testing with the mask is kept, with smaller values indicating more signal being kept.

(35) The first hyper-parameter NN is a trained neural network that has been machine trained with training data including training speech (including downstream task labels), the mask (enhancement mask), and parameters of the downstream task NN so as to calculate the optimal first hyper-parameter when noisy speech data is input.

(36) Specifically, the first hyper-parameter NN learning unit **20** learns a weighting factor of the neural network for a neural network including an input layer that accepts inputs of speech data and an output layer that outputs the first hyper-parameter to minimize a loss function indicating an error between the downstream task label and a processing result of the downstream task processing unit **32** (downstream task NN), which will be described below, using the training data including training speech (including downstream task labels) with noise, the mask and parameters of the downstream

task NN.

(37) The content of the loss function depends on the manner of the downstream task. For example, it is assumed that the downstream task is speaker recognition and the downstream task processing unit **32** outputs the posterior of the estimated speaker ID as a processing result. In this case, the first hyper-parameter NN learning unit **20** may learn a weighting factor of the neural network to minimize the cross-entropy error between the ground truth speaker ID indicated by the downstream task label and the estimated output speaker ID posterior of the estimated speaker ID.

(38) The first hyper-parameter NN parameter storage unit **22** stores the parameters of the first hyper-parameter NN learned by the first hyper-parameter NN learning unit **20**.

(39) The second speech enhancement unit **24** determines the enhancement mask from the test utterance, when the test utterance is inputted, using the speech enhancement NN parameters stored in the speech enhancement NN parameter storage unit **14**. The method of determining the mask is the same as the method performed by the first speech enhancement unit **16**. Since the second speech enhancement unit **24** obtains the speech enhancement NN parameter stored in the speech enhancement NN parameter storage unit **14**, the second speech enhancement unit **24** can also be said to have a speech enhancement NN.

(40) The first hyper-parameter optimization unit **26** applies, when the test utterance is inputted, the inputted test utterance to the first hyper-parameter NN to calculate the optimized hyper-parameter $[\gamma]$ (i.e., the first hyper-parameter).

(41) The mask generation unit **28** generates a mask (i.e., an adaptive mask) $M_{sup}[\gamma]$ from the enhancement mask determined by the second speech enhancement unit **24** and the first hyper-parameter $[\gamma]$ optimized by the first hyper-parameter optimization unit **26** to enhance the test utterances adapted to the downstream task. Specifically, the mask generation unit **28** generates an adaptive mask with the first hyper-parameter $[\gamma]$ as a power of the mask. The adaptive mask $M_{sup}[\gamma]$ is also a time-frequency matrix of real values.

(42) The adapted speech enhancement unit **30** applies the adaptive mask $M_{sup}[\gamma]$ to the test utterances to generate the enhanced speech data (hereinafter referred to as the adaptive speech data). Note that the enhanced speech data Y' is represented by Equation 1 shown below, where Y is the test utterance.

$$Y' = Y * M \quad (\text{Equation 1})$$

(43) The downstream task processing unit **32** inputs the adaptive speech data generated by the adapted speech enhancement unit **30** into the downstream task NN and outputs the processing result. The manner of the downstream task NN is determined according to the processing content. For example, if the content of the downstream task is speaker recognition, the downstream task processing unit **32** may output a posterior of the speaker ID as a processing result, as described above.

(44) The first hyper-parameter NN learning unit **20** calculates an error using a loss function based on the output processing result, and propagates the calculated error to the first hyper-parameter NN.

(45) The training speech input unit **12**, the first speech enhancement unit **16**, the first hyper-parameter NN training unit **20**, the second speech enhancement unit **24**, the first hyper-parameter optimization unit **26**, the mask generation unit **28**, the adapted speech enhancement unit **30**, and a downstream task processing unit **32** are implemented by a CPU of a computer operating according to a program (hyper-parameter optimization program). For example, the program may be stored in a storage medium (not shown) provided by the parameter optimization system **100**, with the CPU reading the program and, according to the program, operating as the training speech input unit **12**, the first speech enhancement unit **16**, the first hyper-parameter NN training unit **20**, the second speech enhancement unit **24**, the first hyper-parameter optimization unit **26**, the mask generation unit **28**, the adapted speech enhancement unit **30**, and a downstream task processing unit **32**. The functions of the parameter optimization system **100** may be provided in the form of SaaS (Software as a Service).

(46) The training speech input unit **12**, the first speech enhancement unit **16**, the first hyper-parameter NN training unit **20**, the second speech enhancement unit **24**, the first hyper-parameter optimization unit **26**, the mask generation unit **28**, the adapted speech enhancement unit **30**, and a downstream task processing unit **32** may each be implemented by dedicated hardware. All or part of the components of each device may be implemented by general-purpose or dedicated circuitry, processors, or combinations thereof. They may be configured with a single chip, or configured with a plurality of chips connected via a bus. All or part of the components of each device may be implemented by a combination of the above-mentioned circuitry or the like and program.

(47) In the case where all or part of the components of each device is implemented by a plurality of information processing devices, circuitry, or the like, the plurality of information processing devices, circuitry, or the like may be centralized or distributed. For example, the information processing devices, circuitry, or the like may be implemented in a form in which they are connected via a communication network, such as a client-and-server system or a cloud computing system.

(48) The speech enhancement NN parameter storage unit **14**, the downstream task NN parameter storage unit **18**, and the first hyper-parameter NN parameter storage unit **22** are realized, for example, by a magnetic disk or the like.

(49) Next, an operation example of the parameter optimization system according to the present exemplary embodiment will be described. FIG. 2 depicts a flowchart illustrating the process of the first exemplary embodiment of the parameter optimization system **100** according to the disclosure.

(50) The second speech enhancement unit **24** inputs the test utterance into the speech enhancement NN to determine the enhancement mask (step **S11**). The first hyper-parameter optimization unit **26** inputs the test utterance into the first hyper-parameter NN and outputs the first hyper-parameter $[\gamma]$ (step **S12**). Then, the mask generation unit **28** generates the adaptive mask $M_{sup.}[\gamma]$ from the determined enhancement mask and the first hyper-parameter (step **S13**).

(51) Thereafter, the adapted speech enhancement unit **30** generates adaptive speech data from the test utterances using the adaptive mask $M_{sup.}[\gamma]$, and the downstream task processing unit **32** inputs the generated adaptive speech data into the downstream task NN and outputs the processing result.

(52) In this exemplary embodiment, the first hyper-parameter NN training unit **20** learns the first hyper-parameter NN by a machine learning using data including the training speech, the enhancement mask, and the parameters of the neural network of the downstream task as training data.

(53) As described above, in the present exemplary embodiment, when the test utterance is input, the second speech enhancement unit **24** determines the enhancement mask generated based on the mask for speech enhancement from the test utterance, and the first hyper-parameter optimization unit **26** determines the first hyper-parameter $[\gamma]$ when the test utterance is input. Then, the mask generation unit **28** generates the adaptive mask $M_{sup.}[\gamma]$ in which the first hyper-parameter is a power of the mask. Thus, an optimal hyper-parameter indicating the power of the mask for speech enhancement can be determined according to the nature of the downstream task.

(54) In other words, in this exemplary embodiment, an adaptive mask is generated from the first hyper-parameter NN learned by the first hyper-parameter NN learning unit **20** to enhance the test utterance adapted to the downstream task. As a result, the speech can be enhanced taking into account the trade-off between speech clarity and processing accuracy for the downstream tasks speech.

Second Exemplary Embodiment

(55) Next, a second exemplary embodiment of the hyper-parameter optimization system of the disclosure will be described. In the first exemplary embodiment, the configuration in which the parameters of the downstream task NN are learned beforehand and stored in the downstream task NN parameter storage unit **18** is described as an example. The second exemplary embodiment describes an example configuration in which the first hyper-parameter NN and the downstream

task NN are learned together.

(56) FIG. 3 depicts an exemplary block diagram illustrating the structure of a second exemplary embodiment of a hyper-parameter optimization system according to the disclosure. The hyper-parameter optimization system **200** according to the second exemplary embodiment includes the training speech input unit **12**, the speech enhancement NN parameter storage unit **14**, the first speech enhancement unit **16**, a downstream task label storage unit **34**, a first hyper-parameter NN and downstream task NN training unit **36**, the downstream task NN parameter storage unit **18**, the first hyper-parameter NN parameter storage unit **22**, the second speech enhancement unit **24**, the first hyper-parameter optimization unit **26**, the mask generation unit **28**, the adapted speech enhancement unit **30**, and a downstream task processing unit **32**.

(57) That is, the hyper-parameter optimization system **200** of this exemplary embodiment differs from the hyper-parameter optimization system **100** of the first exemplary embodiment in that it further provides a downstream task label storage unit **34** and has a first hyper-parameter NN and downstream task NN learning unit **36** instead of the first hyper-parameter NN learning unit **20**. Otherwise, the configuration is the same as in the first exemplary embodiment.

(58) The downstream task label storage unit **34** stores the task training data used by the first hyper-parameter NN and downstream task NN learning unit **36** for learning the downstream task NN, which will be described below. The task training data is data associated with the speech data and a correct label for the downstream task (i.e., downstream task label), and is defined according to the content of the downstream task. For example, if the downstream task is speaker recognition, the downstream task label storage unit **34** may store the data corresponding to the noiseless speech (Hereinafter referred to as clean speech) and the speaker ID as task training data. Further, for example, if the downstream task is speech recognition, the downstream task label storage unit **34** may store data corresponding to the clean speech and text contents as the task training data.

(59) The first hyper-parameter NN and downstream task NN learning unit **36** learns the first hyper-parameter NN and the downstream task NN. Specifically, the first hyper-parameter NN and downstream task NN learning unit **36** learns the weighting factors of the first hyper-parameter NN including an input layer that accepts inputs of speech data and an output layer that outputs the first hyper-parameter, and the downstream task NN including an input layer that accepts inputs of speech data and an output layer that outputs a result of processing by the downstream task so as to minimize a loss function indicating an error between the downstream task label and the processing result of the downstream task processing unit **32**, using the training speech, the mask and the task training data as training data for the first hyper-parameter NN and the downstream task NN.

(60) The training speech input unit **12**, the first speech enhancement unit **16**, the first hyper-parameter NN and downstream task NN learning unit **36**, the second speech enhancement unit **24**, the first hyper-parameter optimization unit **26**, the mask generation unit **28**, the adapted speech enhancement unit **30**, and a downstream task processing unit **32** are implemented by a CPU of a computer operating according to a program (hyper-parameter optimization program).

(61) Next, an operation example of the parameter optimization system according to the present exemplary embodiment will be described. FIG. 4 depicts a flowchart illustrating the process of the second exemplary embodiment of the parameter optimization system **200** according to the disclosure.

(62) The first hyper-parameter NN and downstream task NN learning unit **36** learns the first hyper-parameter NN and the downstream task NN by a machine learning using the training speech, the mask, and the task training data as training data (step **S21**). Thereafter, the processes of steps **S11** to **S13** in FIG. 2 are performed using the learned first hyper-parameter NN and the downstream task NN.

(63) As described above, in the present exemplary embodiment, the first hyper-parameter NN and the downstream task NN learning unit **36** learns the first hyper-parameter NN and the downstream task NN. Therefore, in addition to the effect of the first exemplary embodiment, the downstream

task NN can be optimized at the same time.

Third Exemplary Embodiment

(64) Next, a third exemplary embodiment of the hyper-parameter optimization system of the disclosure will be described. In the first exemplary embodiment and the second exemplary embodiment, the configuration in which the parameters of the speech enhancement NN are learned beforehand and stored in the speech enhancement NN parameter storage unit **14** is described as an example. The third exemplary embodiment describes an example configuration in which the first hyper-parameter NN and the speech enhancement NN are learned together.

(65) FIG. 5 depicts an exemplary block diagram illustrating the structure of a third exemplary embodiment of a hyper-parameter optimization system according to the disclosure. The hyper-parameter optimization system **300** according to the third exemplary embodiment includes a noise storage unit **42**, a clean speech storage unit **44**, a combination unit **46**, a noisy speech storage unit **48**, a second hyper-parameter optimization unit **50**, a target calculation unit **52**, a target storage unit **54**, and a first hyper-parameter NN and speech enhancement NN learning unit **56**.

(66) Furthermore, the hyper-parameter optimization system **300** of the third exemplary embodiment includes the speech enhancement NN parameter storage unit **14**, a downstream task label storage unit **34**, the downstream task NN parameter storage unit **18**, the first hyper-parameter NN parameter storage unit **22**, the second speech enhancement unit **24**, the first hyper-parameter optimization unit **26**, the mask generation unit **28**, the adapted speech enhancement unit **30**, and a downstream task processing unit **32** as a similar configuration to the hyper-parameter optimization system **200** of the second embodiment.

(67) The noise storage unit **42** stores one or more types of noise signals assumed for the test utterances. The clean speech storage unit **44** stores a noiseless speech (clean speech) that may be acquired under the same conditions (similar domain) and so on as the test utterance. The noise signal and the clean speech are prepared in advance by the user or others and stored in the noise storage unit **42** and the clean speech storage unit **44**, respectively.

(68) The combination unit **46** combines the noise signal and the clean speech to generate a speech with noise (hereinafter sometimes referred to as noisy speech). The generated noisy speech is represented, for example, by the following equation. The x here is used to determine the SNR (Speech-noise ratio:speech-noise ratio) of the generated noisy speech. Since the method of generating the noisy speech is widely known, a detailed explanation is omitted here.

Noisy speech=(noise signal $\times$ x +clean speech)

(69) The combination unit **46** stores the generated noisy speech in the noisy speech storage unit **48**.

(70) The noisy speech storage unit **48** stores the noisy speech. The noisy speech storage unit **48** may store the noisy speech generated by the combination unit **46** and may store multi-SNR training data.

(71) The second hyper-parameter optimization unit **50** determines a hyper-parameter $[\alpha]$ which indicates the degree to which the speech enhancement NN keeps the speech (in other words, the degree to which the noise is removed). In the following description, this hyper-parameter $[\alpha]$ will be referred to as the second hyper-parameter.

(72) More specifically, the second hyper-parameter $[\alpha]$ is a hyper-parameter that controls in training how much weight the speech enhancement NN sets to keep the speech and how much weight the speech enhancement NN sets to denoise in speech enhancement using the mask. The second hyper-parameter $[\alpha]$ is a positive scalar value.

(73) In this exemplary embodiment, the second hyper-parameter is a predetermined hyper-parameter that is manually tuned by the user or others. The second hyper-parameter optimization unit **50** determines that this hyper-parameter is used as the second hyper-parameter $[\alpha]$. The second hyper-parameter may be, for example, a value optimized based on gradient methods.

(74) The target calculation unit **52** calculates $M.\sup.[\alpha]$ from the mask and the second hyper-parameter $[\alpha]$ predetermined for speech enhancement by a power of the second hyper-

parameter [alpha] of the mask. $M_{sup}[\alpha]$ may be a matrix indicating the degree of speech enhancement calculated based on the mask, and this $M_{sup}[\alpha]$ may be referred to as “target”. $M_{sup}[\alpha]$ is a time-frequency matrix of real values, as well as the mask M and $M_{sup}[\gamma]$.

(75) The target storage unit **54** stores the target $M_{sup}[\alpha]$ calculated by the target calculation unit **52**.

(76) The first hyper-parameter NN and speech enhancement NN learning unit **56** learns the first hyper-parameter NN and the speech enhancement NN. Specifically, using the data including noisy speech, the target, task training data, and parameters of the downstream task NN as training data, the first hyper-parameter NN and speech enhancement NN learning unit **56** learns the weighting factors of the first hyper-parameter NN (including an input layer that accepts inputs of speech data and an output layer that outputs the first hyper-parameter) and the weighting factors of the speech enhancement NN (including an input layer that accepts inputs of speech data and an output layer that outputs the target) so as to minimize the weighted sum of a first loss indicating the error between the downstream task label and the processing result of the downstream task and a second loss indicating the error between the target included in the training data and the target output by the speech enhancement NN.

(77) The combination unit **46**, the second hyper-parameter optimization unit **50**, the target calculation unit **52**, the first hyper-parameter NN and speech enhancement NN learning unit **56**, the second speech enhancement unit **24**, the first hyper-parameter optimization unit **26**, the mask generation unit **28**, the adapted speech enhancement unit **30**, and a downstream task processing unit **32** are implemented by a CPU of a computer operating according to a program (hyper-parameter optimization program).

(78) Next, an operation example of the parameter optimization system according to the present exemplary embodiment will be described. FIG. 6 depicts a flowchart illustrating the process of the third exemplary embodiment of the parameter optimization system **300** according to the disclosure.

(79) The first hyper-parameter NN and speech enhancement NN learning unit **56** learns the first hyper-parameter NN and the speech enhancement NN by a machine learning using the noisy speech generated by the combination unit **46**, the target calculated by the target calculation unit **52**, the task training data, and the parameters of the downstream task NN as training data (step S31). Thereafter, the processes of steps S11 to S13 in FIG. 2 are performed using the learned first hyper-parameter NN and the speech enhancement NN.

(80) As described above, in the present exemplary embodiment, the first hyper-parameter NN and speech enhancement NN learning unit **56** learns the first hyper-parameter NN and the speech enhancement NN. Therefore, in addition to the effect of the first exemplary embodiment, the speech enhancement NN can be optimized at the same time.

Fourth Exemplary Embodiment

(81) Next, a fourth exemplary embodiment of the hyper-parameter optimization system of the disclosure will be described. The fourth exemplary embodiment describes an example of a configuration in which the first hyper-parameter NN, the speech enhancement NN, and the downstream task NN are learned together.

(82) FIG. 7 depicts an exemplary block diagram illustrating the structure of a fourth exemplary embodiment of a hyper-parameter optimization system according to the disclosure. Compared with the configuration of the hyper-parameter optimization system **300** of the third embodiment, the hyper-parameter optimization system **400** of this exemplary embodiment includes a first hyper-parameter NN and downstream task NN and speech enhancement NN learning unit **62** instead of a first hyper-parameter NN and speech enhancement NN learning unit **56**. Otherwise, the configuration is the same as in the third exemplary embodiment.

(83) The first hyper-parameter NN and downstream task NN and speech enhancement NN learning unit **62** learns the first hyper-parameter NN, downstream task NN, and the speech enhancement

NN. Specifically, using the data including noisy speech, the target, and task training data as training data, the first hyper-parameter NN and downstream task NN and speech enhancement NN learning unit **62** learns the weighting factors of the first hyper-parameter NN (including an input layer that accepts inputs of speech data and an output layer that outputs the first hyper-parameter), the weighting factors of the downstream task NN (including an input layer that accepts inputs of speech data and an output layer that outputs the processing result of the downstream task), and the weighting factors of the speech enhancement NN (including an input layer that accepts inputs of speech data and an output layer that outputs the target) so as to minimize the weighted sum of a first loss indicating the error between the downstream task label and the processing result of the downstream task and a second loss indicating the error between the target included in the training data and the target output by the speech enhancement NN.

(84) The combination unit **46**, the second hyper-parameter optimization unit **50**, the target calculation unit **52**, the first hyper-parameter NN and downstream task NN and speech enhancement NN learning unit **62**, the second speech enhancement unit **24**, the first hyper-parameter optimization unit **26**, the mask generation unit **28**, the adapted speech enhancement unit **30**, and a downstream task processing unit **32** are implemented by a CPU of a computer operating according to a program (hyper-parameter optimization program).

(85) Next, an operation example of the parameter optimization system according to the present exemplary embodiment will be described. FIG. **8** depicts a flowchart illustrating the process of the fourth exemplary embodiment of the parameter optimization system **400** according to the disclosure.

(86) The first hyper-parameter NN and downstream task NN and speech enhancement NN learning unit **62** learns the first hyper-parameter NN, the speech enhancement NN, and the downstream task NN by a machine learning using the noisy speech, the target, and the task training data as training data (step **S41**). Thereafter, the processes of steps **S11** to **S13** in FIG. **2** are performed using the learned first hyper-parameter NN, the speech enhancement NN, and the downstream task NN.

(87) As described above, in the present exemplary embodiment, the first hyper-parameter NN and downstream task NN and speech enhancement NN learning unit **62** learns the first hyper-parameter NN, the downstream task NN, and the speech enhancement NN. Thus, in addition to the effect of the first embodiment, the speech enhancement NN and the downstream task NN can be optimized simultaneously.

(88) Next, an outline of the disclosure will be described. FIG. **9** depicts a block diagram illustrating an outline of the hyper-parameter optimization system according to the disclosure. The hyper-parameter optimization system **80** (for example, hyper-parameter optimization system **100** to **400**) including: a speech enhancement means **81** (e.g., the second speech enhancement unit **24**) which determines an enhancement mask generated based on a mask (e.g., M) for speech enhancement, when a test utterance is input as speech data; a first hyper-parameter optimization means **82** (e.g., the first hyper-parameter optimization unit **26**) which determines, when the test utterance is input, a first hyper-parameter (e.g., $[\gamma]$) which is a hyper-parameter representing the degree to which the signal representing the test utterance is kept using the mask, and the first hyper-parameter which is set to take into account a downstream task that is processed using an enhanced test utterance; and a mask generation means **83** (e.g., the mask generation unit **28**) which generates an adaptive mask (e.g., $M.\text{sup}.[\gamma]$) from the determined enhancement mask and the first hyper-parameter that enhances the test utterance for the downstream task.

(89) The mask generation means **83** generates the adaptive mask in which the first hyper-parameter is a power of the mask.

(90) With such a configuration, it is possible to determine the optimum hyper-parameters of the mask that enhances speech according to the nature of the downstream task.

(91) The first hyper-parameter optimization means may have a trained first hyper-parameter neural network that has been machine learning using training data including downstream task label

indicating a processing result of the downstream task, training speech with noise, the enhancement mask, and parameters of a neural network of the downstream task, so as to output the first hyper-parameter when speech data with noise is input.

(92) The speech enhancement means **81** may have a trained speech enhancement neural network that has been machine learning so as to output the enhancement mask from speech data when the speech data with noise is input.

(93) The hyper-parameter optimization system **80** may include a first hyper-parameter neural network training means (e.g., the first hyper-parameter NN training unit **20**) which trains a first hyper-parameter neural network, which includes an input layer that accepts speech data and an output layer that outputs the first hyper-parameter, by a machine learning using training data including training speech, the enhancement mask, and parameters of a neural network of the downstream task.

(94) The hyper-parameter optimization system **80** may include a first hyper-parameter neural network and downstream task neural network training means (e.g., first hyper-parameter NN and downstream task NN training unit **36**) which trains a first hyper-parameter neural network and a downstream task neural network, the first hyper-parameter neural network which includes an input layer that accepts speech data and an output layer that outputs the first hyper-parameter, the downstream task neural network which includes an input layer that accepts speech data and an output layer that outputs results of the downstream task, by a machine learning using training data including training speech, the enhancement mask, and task training data associated with speech data and a correct label for downstream task.

(95) The hyper-parameter optimization system **80** may include a first hyper-parameter neural network and speech enhancement neural network training means (e.g., first hyper-parameter NN and speech enhancement NN learning unit **56**) which trains a first hyper-parameter neural network and a speech enhancement neural network, the first hyper-parameter neural network which includes an input layer that accepts speech data and an output layer that outputs the first hyper-parameter, the speech enhancement neural network which includes an input layer that accepts speech data and an output layer that outputs a target indicating the degree of speech enhancement calculated based on the mask, by a machine learning using training data including speech data with noise, the target, task training data associated with speech data and a correct label for downstream task, and parameters of a neural network of the downstream task.

(96) The hyper-parameter optimization system **80** may include a three types of neural network learning means (e.g., first hyper-parameter NN and downstream task NN and speech enhancement NN learning unit **62**) which trains a first hyper-parameter neural network, a downstream task neural network, and a speech enhancement neural network, the first hyper-parameter neural network which includes an input layer that accepts speech data and an output layer that outputs the first hyper-parameter, the speech enhancement neural network which includes an input layer that accepts speech data and an output layer that outputs a target indicating the degree of speech enhancement calculated based on the mask, and the downstream task neural network which includes an input layer that accepts speech data and an output layer that outputs results of the downstream task, by a machine learning using training data including speech data with noise, the target, and task training data associated with speech data and correct labels for downstream task.

(97) The hyper-parameter optimization system **80** may include: a second hyper-parameter optimization means (e.g., second hyper-parameter optimization unit **50**) that optimizes a second hyper-parameter that is used for training of the speech enhancement neural network; and a target calculating means (e.g., target calculation unit **52**) that receives the second hyper-parameter from the second hyper-parameter optimization means and calculates a mask (e.g., $M_{sup}[\alpha]$) in which the second hyper-parameter is a power of the mask as the target.

(98) The second hyper-parameter may be optimized based on at least one of gradient methods.

(99) The mask may be a matrix consisting of real or complex continuous values taking at least one

form of Ideal Ratio Mask, Complex Ideal Ratio Mask, Spectral Magnitude Mask, and Phase-Sensitive Mask.

(100) The hyper-parameter optimization system **80** may include: an adapted speech enhancement means (e.g., adapted speech enhancement unit **30**) which applies the adaptive mask to the test utterance to generate adaptive speech data which is enhanced speech data; and a downstream task processing means (e.g., downstream task processing unit **32**) which inputs the adaptive speech data and outputs a processing result.

(101) FIG. **10** depicts a schematic block diagram illustrating a configuration of a computer according to at least one of the exemplary embodiments. A computer **1000** includes a CPU **1001**, a main storage device **1002**, an auxiliary storage device **1003**, and an interface **1004**.

(102) Each of the above-described hyper-parameter parameter optimization system is mounted on the computer **1000**. The operation of the respective processing units described above is stored in the auxiliary storage device **1003** in the form of a program (a hyper-parameter parameter optimization program). The CPU **1001** reads the program from the auxiliary storage device **1003**, deploys the program in the main storage device **1002**, and executes the above processing according to the program.

(103) Note that at least in one of the exemplary embodiments, the auxiliary storage device **1003** is an exemplary non-transitory physical medium. Other examples of non-transitory physical medium include a magnetic disc, a magneto-optical disk, a CD-ROM, a DVD-ROM, and a semiconductor memory that are connected via the interface **1004**. In the case where the program is distributed to the computer **1000** by a communication line, the computer **1000** distributed with the program may deploy the program in the main storage device **1002** to execute the processing described above.

(104) Incidentally, the program may implement a part of the functions described above. The program may implement the aforementioned functions in combination with another program stored in the auxiliary storage device **1003** in advance, that is, the program may be a differential file (differential program).

(105) While the invention has been particularly shown and described with reference to example embodiments thereof, the invention is not limited to these embodiments. It will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope of the disclosure as defined by the claims.

(106) The whole or part of the example embodiments disclosed above can be described as, but not limited to, the following supplementary notes. (Supplementary note 1) A hyper-parameter optimization system comprising: a speech enhancement means which determines an enhancement mask generated based on a mask for speech enhancement, when a test utterance is input as speech data; a first hyper-parameter optimization means which determines, when the test utterance is input, a first hyper-parameter which is a hyper-parameter representing the degree to which the signal representing the test utterance is kept using the mask, and the first hyper-parameter which is set to take into account a downstream task that is processed using an enhanced test utterance; and a mask generation means which generates an adaptive mask from the determined enhancement mask and the first hyper-parameter that enhances the test utterance for the downstream task, wherein the mask generation means generates the adaptive mask in which the first hyper-parameter is a power of the mask. (Supplementary note 2) The hyper-parameter optimization system according to Supplementary note 1, wherein the first hyper-parameter optimization means has a trained first hyper-parameter neural network that has been machine learning using training data including downstream task label indicating a processing result of the downstream task, training speech with noise, the enhancement mask, and parameters of a neural network of the downstream task, so as to output the first hyper-parameter when speech data with noise is input. (Supplementary note 3) The hyper-parameter optimization system according to Supplementary note 1 or 2, wherein the speech enhancement means has a trained speech enhancement neural network that has been machine learning so as to output the enhancement mask from speech data when the speech data with noise is

input. (Supplementary note 4) The hyper-parameter optimization system according to any one of Supplementary notes 1 to 3, further comprising a first hyper-parameter neural network training means which trains a first hyper-parameter neural network, which includes an input layer that accepts speech data and an output layer that outputs the first hyper-parameter, by a machine learning using training data including training speech, the enhancement mask, and parameters of a neural network of the downstream task. (Supplementary note 5) The hyper-parameter optimization system according to any one of Supplementary notes 1 to 3, further comprising a first hyper-parameter neural network and downstream task neural network training means which trains a first hyper-parameter neural network and a downstream task neural network, the first hyper-parameter neural network which includes an input layer that accepts speech data and an output layer that outputs the first hyper-parameter, the downstream task neural network which includes an input layer that accepts speech data and an output layer that outputs results of the downstream task, by a machine learning using training data including training speech, the enhancement mask, and task training data associated with speech data and a correct label for downstream task. (Supplementary note 6) The hyper-parameter optimization system according to any one of Supplementary notes 1 to 3, further comprising a first hyper-parameter neural network and speech enhancement neural network training means which trains a first hyper-parameter neural network and a speech enhancement neural network, the first hyper-parameter neural network which includes an input layer that accepts speech data and an output layer that outputs the first hyper-parameter, the speech enhancement neural network which includes an input layer that accepts speech data and an output layer that outputs a target indicating the degree of speech enhancement calculated based on the mask, by a machine learning using training data including speech data with noise, the target, task training data associated with speech data and a correct label for downstream task, and parameters of a neural network of the downstream task. (Supplementary note 7) The hyper-parameter optimization system according to any one of Supplementary notes 1 to 3, further comprising a three types of neural network learning means which trains a first hyper-parameter neural network, a downstream task neural network, and a speech enhancement neural network, the first hyper-parameter neural network which includes an input layer that accepts speech data and an output layer that outputs the first hyper-parameter, the speech enhancement neural network which includes an input layer that accepts speech data and an output layer that outputs a target indicating the degree of speech enhancement calculated based on the mask, and the downstream task neural network which includes an input layer that accepts speech data and an output layer that outputs results of the downstream task, by a machine learning using training data including speech data with noise, the target, and task training data associated with speech data and correct labels for downstream task. (Supplementary note 8) The hyper-parameter optimization system according to Supplementary note 6 or 7, further comprising: a second hyper-parameter optimization means that optimizes a second hyper-parameter that is used for training of the speech enhancement neural network; and a target calculating means that receives the second hyper-parameter from the second hyper-parameter optimization means and calculates a mask in which the second hyper-parameter is a power of the mask as the target. (Supplementary note 9) The hyper-parameter optimization system according to Supplementary note 8, wherein the second hyper-parameter is optimized based on at least one of gradient methods. (Supplementary note 10) The hyper-parameter optimization system according to any one of claims 1 to 9, wherein, the mask is a matrix consisting of real or complex continuous values taking at least one form of Ideal Ratio Mask, Complex Ideal Ratio Mask, Spectral Magnitude Mask, and Phase-Sensitive Mask. (Supplementary note 11) The hyper-parameter optimization system according to any one of Supplementary notes 1 to 10, further comprising: an adapted speech enhancement means which applies the adaptive mask to the test utterance to generate adaptive speech data which is enhanced speech data; and a downstream task processing means which inputs the adaptive speech data and outputs a processing result. (Supplementary note 12) A hyper-parameter optimization method comprising: determining an

enhancement mask generated based on a mask for speech enhancement, when a test utterance is input as speech data; determining, when the test utterance is input, a first hyper-parameter which is a hyper-parameter representing the degree to which the signal representing the test utterance is kept using the mask, and the first hyper-parameter which is set to take into account a downstream task that is processed using an enhanced test utterance; and generating an adaptive mask from the determined enhancement mask and the first hyper-parameter that enhances the test utterance for the downstream task, wherein the adaptive mask in which the first hyper-parameter is a power of the mask is generated. (Supplementary note 13) A non-transitory computer readable information recording medium storing a hyper-parameter optimization program, when executed by a processor, that performs a method for: determining an enhancement mask generated based on a mask for speech enhancement, when a test utterance is input as speech data; determining, when the test utterance is input, a first hyper-parameter which is a hyper-parameter representing the degree to which the signal representing the test utterance is kept using the mask, and the first hyper-parameter which is set to take into account a downstream task that is processed using an enhanced test utterance; and generating an adaptive mask from the determined enhancement mask and the first hyper-parameter that enhances the test utterance for the downstream task, wherein the adaptive mask in which the first hyper-parameter is a power of the mask is generated. (Supplementary note 14) A hyper-parameter optimization program for causing a computer to execute: a speech enhancement process of determining an enhancement mask generated based on a mask for speech enhancement, when a test utterance is input as speech data; a first hyper-parameter optimization process of determining, when the test utterance is input, a first hyper-parameter which is a hyper-parameter representing the degree to which the signal representing the test utterance is kept using the mask, and the first hyper-parameter which is set to take into account a downstream task that is processed using an enhanced test utterance; and a mask generation process of generating an adaptive mask from the determined enhancement mask and the first hyper-parameter that enhances the test utterance for the downstream task, wherein the adaptive mask in which the first hyper-parameter is a power of the mask is generated.

REFERENCE SIGNS LIST

(107) **12** training speech input unit **14** speech enhancement neural network parameter storage unit **16** first speech enhancement unit **18** downstream task neural network parameter storage unit **20** first hyper-parameter neural network training unit **22** first hyper-parameter NN storage unit **24** second speech enhancement unit **26** first hyper-parameter optimization unit **28** mask generation unit **30** adaptive speech enhancement unit **32** downstream task processing unit **34** downstream task label storage unit **36** first hyper-parameter NN and downstream task NN training unit **42** noise storage unit **44** clean speech storage unit **46** combination unit **48** noisy speech storage unit **50** second hyper-parameter optimization unit **52** target calculation unit **54** target storage unit **56** first hyper-parameter NN and speech enhancement NN learning unit **62** speech enhancement NN and first hyper-parameter NN and downstream task NN learning unit

Claims

1. A hyper-parameter optimization system comprising: a memory storing instructions; and one or more processors configured to execute the instructions to: determine an enhancement mask generated based on a mask for speech enhancement, in a case where a test utterance is input as speech data; determine, in the case where the test utterance is input, a first hyper-parameter which is a hyper-parameter representing a degree to which a signal representing the test utterance is kept using the mask, and the first hyper-parameter which is set to take into account a downstream task that is processed using an enhanced test utterance; generate an adaptive mask from the determined enhancement mask and the first hyper-parameter that enhances the test utterance for the downstream task, and in which the first hyper-parameter is a power of the mask; and train a first

hyper-parameter neural network, a downstream task neural network, and a speech enhancement neural network, the first hyper-parameter neural network which includes an input layer that accepts the speech data and an output layer that outputs the first hyper-parameter, the speech enhancement neural network which includes an input layer that accepts the speech data and an output layer that outputs a target indicating a degree of the speech enhancement calculated based on the mask, and the downstream task neural network which includes an input layer that accepts the speech data and an output layer that outputs results of the downstream task, by a machine learning using training data including the speech data with noise, the target, and task training data associated with the speech data and correct labels for the downstream task.

2. The hyper-parameter optimization system according to claim 1, wherein the trained first hyper-parameter neural network that has been trained using the training data including a downstream task label indicating a processing result of the downstream task, training speech with the noise, the enhancement mask, and parameters of a neural network of the downstream task, so as to output the first hyper-parameter in a case where the speech data with the noise is input.

3. The hyper-parameter optimization system according to claim 1, wherein the trained speech enhancement neural network that has been trained so as to output the enhancement mask from the speech data in a case where the speech data with the noise is input.

4. The hyper-parameter optimization system according to claim 1, wherein the processor further executes instructions to: optimize a second hyper-parameter that is used for training of the speech enhancement neural network; and receive the second hyper-parameter and calculate the mask in which the second hyper-parameter is the power of the mask as the target.

5. The hyper-parameter optimization system according to claim 4, wherein the second hyper-parameter is optimized based on at least one of gradient methods.

6. The hyper-parameter optimization system according to claim 1, wherein the mask is a matrix consisting of real or complex continuous values taking at least one form of Ideal Ratio Mask, Complex Ideal Ratio Mask, Spectral Magnitude Mask, and Phase-Sensitive Mask.

7. The hyper-parameter optimization system according to claim 1, wherein the processor further executes instructions to: apply the adaptive mask to the test utterance to generate adaptive speech data which is enhanced speech data; and input the adaptive speech data and output a processing result.

8. A hyper-parameter optimization method performed by a computer and comprising: determining an enhancement mask generated based on a mask for speech enhancement, in a case where a test utterance is input as speech data; determining, in the case where the test utterance is input, a first hyper-parameter which is a hyper-parameter representing a degree to which a signal representing the test utterance is kept using the mask, and the first hyper-parameter which is set to take into account a downstream task that is processed using an enhanced test utterance; generating an adaptive mask from the determined enhancement mask and the first hyper-parameter that enhances the test utterance for the downstream task, and in which the first hyper-parameter is a power of the mask; and training a first hyper-parameter neural network, a downstream task neural network, and a speech enhancement neural network, the first hyper-parameter neural network which includes an input layer that accepts the speech data and an output layer that outputs the first hyper-parameter, the speech enhancement neural network which includes an input layer that accepts the speech data and an output layer that outputs a target indicating a degree of the speech enhancement calculated based on the mask, and the downstream task neural network which includes an input layer that accepts the speech data and an output layer that outputs results of the downstream task, by a machine learning using training data including the speech data with noise, the target, and task training data associated with the speech data and correct labels for the downstream task.

9. A non-transitory computer readable information recording medium storing a hyper-parameter optimization program executable by a processor to perform a method comprising: determining an enhancement mask generated based on a mask for speech enhancement, in a case where a test

utterance is input as speech data; determining, in the case where the test utterance is input, a first hyper-parameter which is a hyper-parameter representing a degree to which a signal representing the test utterance is kept using the mask, and the first hyper-parameter which is set to take into account a downstream task that is processed using an enhanced test utterance; generating an adaptive mask from the determined enhancement mask and the first hyper-parameter that enhances the test utterance for the downstream task, and in which the first hyper-parameter is a power of the mask; and training a first hyper-parameter neural network, a downstream task neural network, and a speech enhancement neural network, the first hyper-parameter neural network which includes an input layer that accepts the speech data and an output layer that outputs the first hyper-parameter, the speech enhancement neural network which includes an input layer that accepts the speech data and an output layer that outputs a target indicating a degree of the speech enhancement calculated based on the mask, and the downstream task neural network which includes an input layer that accepts the speech data and an output layer that outputs results of the downstream task, by a machine learning using training data including the speech data with noise, the target, and task training data associated with the speech data and correct labels for the downstream task.
